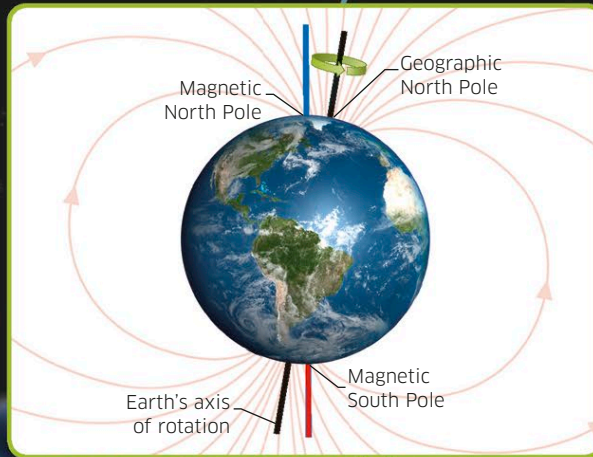


Magnetic and geographic north

There is a difference of a few degrees between the direction that a compass points, known as true north, and the geographic North Pole, which is on the axis of rotation that Earth spins around as it orbits the sun. In reality, the magnetic poles are constantly moving, and reverse completely every few thousand years.



Earth's magnetism

The Earth can be thought of as one big, powerful magnet with a magnetic force field, called the magnetosphere, that stretches thousands of miles into space. The magnetic field is produced by powerful electric currents in the liquid iron and nickel swirling around in Earth's outer core.

Field lines

Representing Earth's magnetic force field, the lines are closest together near the poles, where the field is strongest.

Earth's magnetosphere

The force field extends between 40,000 miles (65,000 km)—around 10 times Earth's radius—and 370,000 miles (600,000 km) into space.